

JEE MAIN REVISION MASTER: SOME BASIC CONCEPTS OF CHEMISTRY SYLLABUS REDUCED

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 40 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Mole Concept & Stoichiometry	31 PYQs	Stoichiometry
High	Concentration Terms (Molarity, Molality, Mole Fraction)	12 PYQs	Concentration

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Mole Concept & Stoichiometry	$n = \frac{w}{M} = \frac{N}{N_A} = \frac{V_{\text{STP}}}{22.4}$ $\text{Combustion: } \text{C}_x\text{H}_y + \left(x + \frac{y}{4}\right)\text{O}_2 \rightarrow x\text{CO}_2 + \frac{y}{2}\text{H}_2\text{O}$
Concentration Terms (Molarity, Molality, Mole Fraction)	$M = \frac{w_B \times 1000}{M_B \times V(\text{mL})}$, $m = \frac{w_B \times 1000}{M_B \times W_A(\text{g})}$, $x_B = \frac{n_B}{n_A + n_B}$, $\text{ppm} = \frac{w_B}{w_{\text{soln}}} \times 10^6$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Mole Concept & Stoichiometry (Very High Yield — 31 PYQs)

Core Principle: Mole relationships, mass-volume stoichiometry, empirical vs molecular formulas, and limiting reagent identification.

■ **Caution / JEE Pitfall:** Always balance the chemical reaction before calculating limiting reagent moles/stoichiometric coefficient ratios.

■ **Speed Trick:** The reactant with the SMALLEST ratio of (Initial Moles / Stoichiometric Coefficient) is the limiting reagent.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Match List - I with List - II. List - I Mass of substance List - II Number of atoms A. 1.8 mg water I. $2 \times 10^{-4} \times N_{\dots}$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "How many grams of residue is obtained by heating 2.76 g of silver carbonate? (Given : Molar mass of C, O and Ag are 12, 16 and 108 g mol^{-1} ..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "The ratio of mass percentage (w/w) of C : H in a hydrocarbon is 12 : 1. It has two carbon atoms. The weight (in g) of CO_2 (g) formed when 3..." [Ans: N/A]

3.2 Concentration Terms (Molarity, Molality, Mole Fraction) (High Yield — 12 PYQs)

Core Principle: Formulas for solution concentrations, temperature dependence, and dilution relation $M_1 V_1 = M_2 V_2$.

■ **Caution / JEE Pitfall:** Molarity (M) and Normality (N) change with temperature (due to volume expansion), while Molality (m) and Mole Fraction (x) are temperature INDEPENDENT.

■ **Speed Trick:** Direct conversion formula: $m = (1000 M / 1000 d - M M_B)$, where d is solution density in g/mL.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "The ratio of mass percentage (w/w) of C : H in a hydrocarbon is 12 : 1. It has two carbon atoms. The weight (in g) of CO₂ (g) formed when 3..." **[Ans: N/A]**
- **[2026 Morning Shift] Q#2 (MCQ):** "What is the mole fraction of water in 10% by weight (w/w) of aqueous urea solution? [Given: Molar mass of H, O, C and N are 1, 16, 12 an..." **[Ans: N/A]**
- **[2025 Evening Shift] Q#2 (MCQ):** "10 mL of 2 M NaOH solution is added to 20 mL of 1 M HCl solution kept in a beaker. Now, 10 mL of this mixture is poured into a volumetric fl..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#3)

An oxide of iron contains 69.9% iron, its empirical formula, is : (Given : Molar mass of Fe and O are 56 and 16 g mol⁻¹ respectively.)

- (A) FeO
- (B) Fe₂O₃ **[Correct]**
- (C) Fe₃O₄
- (D) FeO₃

Correct Answer: B

Solution Analysis: Let the oxide of iron contain 100 g of the compound. Then, Mass of Fe = 69.9 g Mass of O = 100 – 69.9 = 30.1 g Calculate moles of each element Moles of Fe = (69.9 / 56) = 1.248 Moles of O = (30.1 / 16) = 1.881 Find the simplest whole-number ratio Divide each by the smaller number (1.248): Ratio of Fe:O = (1.248 / 1.248) : (1.881 / 1.248) = 1 : 1.51 This ratio is approximately 1 : 1.5 . To remove the fraction, multiply both by 2 : Fe₂O₃ In Fe₂O₃ , Mass of Fe = 2 × 56 = 112 g Mass of O = 3 × 16 = 48 g Total mass = 160 g Percentag

Benchmark Problem 2: JEE Main 2024 (Online) 6th April Evening Shift (Q#1)

Molality (m) of 3M aqueous solution of NaCl is : (Given : Density of solution = 1.25 g mL^{-1} , Molar mass in g mol^{-1} : Na-23, Cl-35.5)

- (A) 2.90 m
- (B) 3.85 m
- (C) 1.90 m
- (D) 2.79 m **[Correct]**

Correct Answer: D

Solution Analysis: To calculate the molality (m) of a 3M (molar) aqueous solution of NaCl , we need to follow certain steps, using the given data and understanding the definitions properly. Molarity (M) is defined as the number of moles of solute per liter of solution. Molality (m) , on the other hand, is defined as the number of moles of solute per kilogram of solvent. The given density of the solution is 1.25 g mL^{-1} which allows us to calculate the mass of the solvent in a given volume of solution. We know: Molarity (M) of NaCl = 3M Density of NaCl solution = 1.25 g mL^{-1} Molar mass of NaCl = Na + Cl